

Lyman's Law, the OCP, and prenasalisation in northern Tōhoku Japanese

Abstract

Modern Tokyo Japanese has a well-known set of morphophonemic alternations known collectively as *rendaku* that involve initial consonants in the second elements of compounds, as in /aki+zora/ 'autumn sky' (cf. /sora/ 'sky'). An alternating element like /sora/~zora/ has an initial voiced obstruent in its *rendaku* allomorph and an initial voiceless obstruent in its non-*rendaku* allomorph, although the paired consonants differ in more than just voicing in most cases. A nearly exceptionless constraint called Lyman's Law blocks *rendaku* in a second element that contains a medial voiced obstruent. The (8th-century) Old Japanese precursor of Lyman's Law, however, seems to have been a ban on prenasalised voiced obstruents in consecutive syllables. Support for this characterisation comes from typology, dialectology, and philology.

1 *Rendaku* and Lyman's Law

Modern Tokyo ("standard") Japanese has a morphophonemic phenomenon called *rendaku* 'sequential voicing'. Many morphemes have one allomorph that begins with a voiceless obstruent and another allomorph that begins with a voiced obstruent, as in (1).

- (1) a. /take/ 'bamboo'
b. /take+yabu/ 'bamboo grove' (cf. /yabu/ 'grove')
c. /sao+dake/ 'pole bamboo' (cf. /sao/ 'pole')

The initial voiced obstruent in the /dake/ allomorph in (1c) is an instance of *rendaku*, and the prototypical environment for *rendaku* is the beginning of the second element in a two-element compound.

Although *rendaku* always pairs a voiceless obstruent with a voiced obstruent, the phonetic difference between the two phonemes in each pair is in many cases more than just the absence vs. presence of voicing (Vance 2016: 3–4). The examples in (2) show all the pairings.

- (2) a. /f/[ɸ]~/b/[b] /furo/ 'bath' /asa+buro/ 'morning bath'
b. /h/[ç]~/b/[b] /hito/ 'person' /tabi+bito/ 'traveler'
c. /h/[h]~/b/[b] /haši/ 'bridge' /iši+baši/ 'stone bridge'
d. /t/[t]~/d/[d] /toši/ 'year' /yaku+doši/ 'unlucky year'
e. /č/[tɕ]~/j/[dʑ] /čikara/ 'strength' /soko+jikara/ 'latent strength'
f. /š/[ɕ]~/j/[dʑ] /šima/ 'stripe' /tate+jima/ 'vertical stripe'
g. /c/[ts]~/z/[(d)z] /cuna/ 'rope' /hana+zuna/ 'nose halter'
h. /s/[s]~/z/[(d)z] /sara/ 'dish' /hai+zara/ 'ashtray'
i. /k/[k]~/g/[g] /kame/ 'turtle' /umi+game/ 'sea turtle'

The historical changes that led to the pairings in (2) are well understood (Vance 2015: 397–398), and the pairings before these changes began will be described below in §6. For the minority of Tokyo speakers who maintain a conservative variety with word-medial, syllable-initial

[ŋ] (Hibiya 1999), the pairing in (1i) is [k]~[ŋ], not [k]~[g]. This additional deviation from straightforward [±vce] will be relevant below in §3.

A constraint known as Lyman's Law (Vance 2015: 402–408) prevents *rendaku* in a two-element compound if the second element (E2) contains a medial voiced obstruent. The examples in (3) illustrate.

- (3) a. /me/ 'eye' + /tama/ 'ball' → /me+dama/ 'eyeball'
b. /hana/ 'flower' + /taba/ 'bunch' → /hana+taba/ 'bouquet'

There are very few exceptions to Lyman's Law in the existing vocabulary (Kindaichi 1976: 5; Martin 1987: 115; Suzuki 2005). It does not seem to matter whether the inhibiting voiced obstruent is in the syllable immediately following the potential *rendaku* site or later in E2 (Lyman 1894: 161–162; Martin 1952: 48). The examples in (4) are typical.¹

- (4) a. /umi/ 'sea' + /suzume/ 'sparrow' → /umi+suzume/ 'murrelet'
b. /ko/ 'child' + /hicuji/ 'sheep' → /ko+hicuji/ 'lamb'

The term “*rendaku*-eligible” can be used to refer to an element that begins with a voiceless obstruent as an independent word. Words that begins with a vowel, with a sonorant consonant, or with a voiced obstruent, like the examples in (5), are not *rendaku*-eligible elements because *rendaku* cannot apply.

- (5) a. /aši/ 'leg' (e.g., /ato+aši/ 'hind leg')
b. /nami/ 'wave' (e.g., /yoko+nami/ 'side wave')
c. /zeni/ 'money' (e.g., /hi+zeni/ 'daily wages')

As Kawahara (2016: 33) notes, some accounts in the literature can easily be misinterpreted to mean that *rendaku* always applies to *rendaku*-eligible E2s unless Lyman's Law would be violated. In fact, however, *rendaku* is intractably irregular. Many phonological, morphological, and semantic factors are known to influence the likelihood of *rendaku*, but there is no overarching generalisation that predicts when *rendaku* occurs and when it does not.

First, some *rendaku*-eligible elements are idiosyncratically immune (Irwin 2016: 104–105), including those in (6).

- (6) a. /kasu/ 'dregs' (e.g., /abura+kasu/ 'oily dregs')
b. /himo/ 'string' (e.g., /kucu+himo/ 'shoelace')

The allomorphs */gasu/ and */bimo/ simply do not exist, despite the fact that neither contains a medial voiced obstruent.

Second, many *rendaku*-eligible elements behave inconsistently as E2s, appearing sometimes with *rendaku* but sometimes without, even when no putative inhibiting factor is relevant. The examples in (7) illustrate.

¹ The results of some psycholinguistic experiments (Vance 1980: 258–259; Ihara & Murata 2006: 21–22) have been consistent with a small “distance effect” in responses to compounds containing nonsense E2s: the more syllables there were between the potential *rendaku* site and the inhibitor voiced obstruent, the more likely participants were to choose responses with *rendaku*. A more recent study by Kawahara (2012), however, using a different methodology, found no such effect. The apparent distance effect in the earlier studies was probably due to some other variable that was uncontrolled (Vance 2015: 403–404).

- (7) a. /či/ ‘blood’ /nama+či/ ‘fresh blood’ (cf. /nama/ ‘fresh’)
 /hana+ji/ ‘nosebleed’ (cf. /hana/ ‘nose’)
 b. /hi/ ‘sun’ /asa+hi/ ‘morning sun’ (cf. /asa/ ‘morning’)
 /niši+bi/ ‘westering sun’ (cf. /niši/ ‘west’)

Both E2s in the compounds in (7) are polysemous (as almost all content morphemes are), and the compounds were chosen so that E2 in each pair of compounds would have the same sense. This precaution is necessary because the probability of *rendaku* in an E2 can differ dramatically depending on its sense (Vance 2015: 433; Irwin 2016a: 104–105).

Third, there are individual compounds, such as those in (8), that can be pronounced either with or without *rendaku*.

- (8) a. /waru+kuči/~waru+guči/ ‘bad mouthing’ (cf. /waru-i/ ‘bad’, /kuči/ ‘mouth’)
 b. /kara+seki/~kara+zeki/ ‘dry cough’ (cf. /kara/ ‘emptiness’, /seki/ ‘cough’)

There are many well-known examples of compounds gaining or losing *rendaku* over time (Vance 2007a: 163), and variability like that in (8) is often symptomatic of a shift in progress.

2 Old Japanese and the strong version of Lyman’s Law

The earliest substantial written documents for Japanese record varieties of 8th-century Old Japanese (OJ). At that time, *rendaku* was already well established, but voiced obstruents were prenasalised and did not occur word-initially in the non-mimetic native vocabulary (Frellesvig 2010: 43; Takayama 2015: 627–629). In addition, the precursor of Lyman’s Law in OJ was markedly different. It appears that *rendaku* was blocked if there was a (prenasalised) voiced obstruent in an adjacent syllable on either side of the target segment. This stricter constraint is known as the strong version of Lyman’s Law (Ramsey & Unger 1972: 287–289). The examples in (9) illustrate.²

- (9) a. ^{OJ}/midu+tori/ ‘waterfowl’ [mĩⁿdutori] (inferred pronunciation)
 cf. ^{OJ}/midu/ ‘water’, ^{OJ}/tori/ ‘bird’
 b. ^{OJ}/suzu+pune/ ‘belled boat’ [sũⁿzupune] (inferred pronunciation)
 cf. ^{OJ}/suzu/ ‘bell’, ^{OJ}/pune/ ‘boat’

There was no (prenasalised) voiced obstruent in ^{OJ}/tori/ (9a) or in ^{OJ}/pune/ (9b), but there was in the last syllable of ^{OJ}/midu/ (9a) and in the last syllable of ^{OJ}/suzu/ (9b). Neither E2 was idiosyncratically immune to *rendaku*, as shown by examples like ^{OJ}/yama+dori/ ‘mountain bird’ and ^{OJ}/opo+bune/ ‘large boat’.

The OJ examples cited above are all attested phonographically, which means that they appear in OJ texts represented in *man’yōgana*, that is, Chinese characters used as phonograms (mostly syllabograms). ^{OJ}/suzu+pune/ (9b) appears only once, written <須儒赴泥>, with each character representing a syllable.³ ^{OJ}/midu+tori/ (9a) appears eight times in total, and while six

² Prenasalisation is transcribed phonetically throughout this paper as nasalisation on the immediately preceding vowel followed by a short nasal stop, but actual realisations in the present-day Kahoku-chō dialect (see §4 below) sometimes lack a measurable nasal stop (Miyashita et al. 2016: 186). Herbert (1986) provides a cross-linguistic survey of the phonology of prenasalised stops.

³ This sole occurrence of ^{OJ}/suzu+pune/ is in poem 51 in *Nihon shoki* (720). The Oxford Corpus of Old Japanese (<http://rcjll.orient.ox.ac.uk/home.html>) makes virtually all OJ texts publicly available in searchable electronic

of these occurrences are written logographically, the remaining two are written phonographically.⁴ It is, of course, inappropriate to use the inferred pronunciation of a logographic attestation as evidence in phonological analysis. In particular, there is no direct evidence for the presence or absence of *rendaku* in a logographically written compound. At the same time, a phonographic representation of a compound is not necessarily an accurate reflection of that compound's actual phonemic form. No existing OJ texts are original; they are copies of copies, sometimes with discrepancies between different copies (Frellesvig 2010: 22), and it is implausible to imagine that the critical editions on which scholars rely are entirely free of transmission errors. Furthermore, some individual *man'yōgana* characters were used inconsistently, sometimes representing a syllable beginning with a voiceless obstruent and sometimes representing a syllable beginning with the corresponding (prenasalised) voiced obstruent. In addition, for some compounds attested phonographically more than once, it appears that a form with *rendaku* and a form without *rendaku* co-existed. Consequently, for a compound attested only once, there is no way to know for sure whether or not it was variable with respect to *rendaku*. Historical linguists working on OJ simply have to keep these philological facts of life in mind and do the best they can with the material that is available. Several other phonographically attested OJ compounds will be cited below, and these examples all need to be interpreted with caution.

The strong version of Lyman's Law clearly does not hold in modern Tokyo Japanese, and examples like those in (10) are abundant.

- (10) a. /fude/ 'writing brush' + /hako/ 'box' → /fude+bako/ 'brush case'
 b. /kaze/ 'cold' + /kusuri/ 'medicine' → /kaze+gusuri/ 'cold medicine'

In fact, although some dictionaries list only /mizu+tori/ as the modern counterpart of (9a), it appears that a majority of younger speakers today prefer /mizu+dori/ (Shioda 2001: 102).

3. Lyman's Law and the OCP

Ever since an influential article by Itô and Mester (1986), it has been popular to interpret Lyman's Law as a manifestation of the Obligatory Contour Principle (OCP). Originally proposed as a prohibition against sequences of identical tones in underlying representations (Leben 2011: 326), the OCP was incorporated into autosegmental phonology as a constraint on the tonal tier (Goldsmith 1990: 309–318). The notion of putting non-tonal features on separate tiers made it possible to treat such features as effectively suprasegmental, and a ban on multiple occurrences of voicing could then be formulated as a subcase of the OCP: OCP-[voice] (Odden 2011: 22).

To make this approach work for Lyman's Law, voicing must be a monovalent feature applicable only to obstruents, the sole class of segments for which voicing is distinctive in Japanese. If the non-distinctive voicing of vowels and sonorant consonants were specified too, any morph with two or more voiced segments in a row, such as /kizu/ 'wound' or /ame/ 'rain', would violate OCP-[voice]. If voicing were treated as a traditional binary feature that specifies voiced obstruents as [+vce] and voiceless obstruents as [–vce], the OCP would not block

form. The Old Japanese Rendaku Database (<http://pj.ninjal.ac.jp/rendaku/en/database/>) provides information on every phonographically attested OJ compound with a *rendaku*-eligible E2, including links to the attestations in the Oxford Corpus.

⁴ The six logographic occurrences of ^{OJ}/midu+tori/ are all written <水鳥>, with each character representing a morpheme. The two phonographic occurrences are both in *Man'yōshū* (ca. 760): <水都等利> in poem 3528 (book 14) and <美豆等利> in poem 4337 (book 20). The character <水> could be used either logographically (to represent ^{OJ}/midu/ 'water') or phonographically (to represent the syllable ^{OJ}/mi/).

the strong version of Lyman’s Law. Both would violate a word-domain version of OCP-[prenasal], however, despite the fact that neither has *rendaku*.

In conservative varieties of modern Tokyo Japanese that have word-medial [ŋ] (< OJ [ᵑg]) instead of [g], [ŋ] blocks *rendaku* just as a voiced obstruent does (Kawahara & Zamma 2016: 25–26).⁷ The examples in (12) illustrate.

- (12) a. /fumi/ ‘letter’ /koi+bumi/ ‘love letter’
 b. /fune/ ‘boat’ /curi+bune/ ‘fishing boat’
 c. /fu[ŋ]u/ ‘blowfish’ /tora+fu[ŋ]u/ ‘tiger blowfish’

E2s with medial [m] (12a) or [n] (12b) do not block *rendaku*. Along with the phonetically opaque pairings of voiceless and voiced obstruents in (1) (§1), the inhibiting power of velar nasals shows that Lyman’s Law in modern Tokyo Japanese has lost its phonetic grounding. It presumably has to be learned as a language-specific constraint.

Furthermore, Kawahara (2008: 324–327), has argued that OCP-[voice] does not seem like a plausible universal constraint in the first place. Cross-linguistically, OCP constraints on consonants seem to target adjacent syllables and features with acoustic correlates that are more “spread out” in time. Such features, which include aspiration (e.g., Grassman’s Law) and prenasalisation, are susceptible to perceptual confusion (Ohala 1981: 189–196, 1993: 249–257) and can lead listeners to attribute their phonetic correlates to a phonological specification on a single segment. The confusion can be resolved by dissimilation (Blevins 2004: 148–149) or prevented from arising by the OCP.

4. Northern Tōhoku Dialects

Japanese dialects spoken in the northern part of the Tōhoku region of northern Honshū preserve the prenasalisation of voiced obstruents. In 2012, fieldwork on *rendaku* in one representative dialect was carried out in Kahoku-chō, a small town in northern Yamagata Prefecture (Miyashita et al. 2016: 180–183). This survey was led by Miyashita, who is a Kahoku-chō native, and the participants were 24 locally born and raised native speakers (13 male, 11 female) who ranged in age from 63 to 94 at the time of recording.

The outcome of *rendaku* in a conservative northern Tōhoku dialect is typically a prenasalised voiced obstruent. The Kahoku-chō survey participants produced many such forms (Miyashita et al. 2016: 183–192), including those in (13). Phonetic transcriptions of Kahoku-chō forms are prefixed with superscript K to avoid confusion with modern Tokyo forms or with inferred OJ forms.

- (13) a. ^K[tɕ] ‘hand’ + ^K[ɸuɰguɰro] ‘bag’ → ^K[tɕᵑbuɰguɰro] ‘glove’
 b. ^K[hama] ‘beach’ + ^K[kuɰri] ‘chestnut’ → ^K[hamãᵑguɰri] ‘clam’

In most northern Tōhoku dialects, ^{OJ}[ᵑg] has shifted to [ŋ], but many older Kahoku-chō speakers retain [ᵑg], as in ^K[hamãᵑguɰri] (13b).

⁷ The question of whether syllable-initial [ŋ] is an allophone of /g/ or a separate phoneme is controversial (Vance 2008: 214–222) and will not be addressed here. Native speakers of Tokyo Japanese who do not have [ŋ] tend to assume that [ŋ] and [g] must be allophones of the same phoneme.

Another salient characteristic of northern Tōhoku dialects is that, in most cases, the consonants corresponding to Tokyo “standard” voiceless stops and voiceless affricates are voiced (but not prenasalised) intervocalically in word-medial position, as shown in (14).⁸

- (14) a. ^K[hada] ‘flag’ cf. Tokyo /hata/ [hata]
 b. ^K[madzi] ‘town’ cf. Tokyo /mači/ [matei]

The expectation is that a counterpart to Lyman’s Law in the Kahoku-chō dialect would involve prenasalised voiced obstruents, and as ^K[tẽ^mbuɣuro] in (13a) shows, a simple voiced obstruent in E2 (the [g] in this example) does not block *rendaku*.

5. Prenasalisation in Adjacent Syllables

Modern Tokyo Japanese has the compound /nabe+buta/ ‘pot lid’, which exhibits *rendaku* (cf. /nabe/ ‘pot’, /futa/ ‘lid’). The corresponding compound is not ordinarily used in the Kahoku-chō dialect. As a result, only 20 of the 24 Kahoku-chō survey participants were able to come up with something like the expected form in response to a picture prompt. Six of these 20 speakers produced a near-standard form with no prenasalisation and no voicing of the stop in the last syllable. Two other speakers had a prenasalised medial stop in E1 but a voiceless medial stop in E2. Another six speakers had a voiced medial stop in E2 but no prenasalisation. These 14 responses are shown in (15).

- (15) E1: ^K[nã^mbɛ] ‘pot’ E2: ^K[ɸɯta] ‘lid’
^K[nabɛbuta] 6 speakers
^K[nã^mbɛbuta] 2 speakers
^K[nabɛbuda] 6 speakers

These 14 productions all display a lack of full integration into the traditional Kahoku-chō dialect’s phonological system, indicating non-nativeness.

The remaining six speakers all had prenasalised [^mb] for the medial consonant in E1 and voiced [d] for the medial consonant in E2, but only one had prenasalisation on the initial consonant of E2, as shown in (16). Furthermore, this sole instance of E2-initial prenasalisation is not entirely unambiguous. These six productions were puzzling.

- (16) a. ^K[nã^mbɛbuda] 5 speakers
 b. ^K[nã^mbɛbuda] 1 speaker (?)

Even the oldest of the Kahoku-chō participants were born too late not to be impacted by the relentless standardisation policy of the Japanese national government, which began in the Meiji period (1868–1912). Like most northern Tōhoku speakers, Kahoku-chō speakers are acutely aware that voiced obstruents corresponding to Tokyo voiceless obstruents and prenasalised obstruents corresponding to Tokyo voiced obstruents are salient and highly stigmatised. Most Kahoku-chō speakers, especially those who are educated, have a tacit understanding of the correspondences between the local dialect and the Tokyo standard. Consequently, they are capable to some degree of converting local forms to standard forms and vice versa. What is hard to understand about the form ^K[nã^mbɛbuda] in (16a) is that it unabashedly contains both stigmatised features but deviates from the correspondence pattern at the *rendaku* site.

The key to unlocking this mystery was provided in a presentation about another northern Tōhoku dialect, spoken in the town of Shizukuishi-chō in Iwate Prefecture (Uwano 2015). In

⁸ It is likely that Old Japanese “voiceless” obstruents, too, were allophonically voiced in intervocalic position (Frellesvig 2010: 34–36).

the Shizukuishi dialect, prenasalised obstruents in consecutive syllables are phonotactically prohibited. For example, consider the Shizukuishi-chō compound in (17), which corresponds to Tokyo /hana+bi/ ‘fireworks’ (cf. /hana/ ‘flower’, /hi/ ‘fire’) and shows the expected prenasalisation at the *rendaku* site. (Phonetic transcriptions of Shizukuishi-chō forms are prefixed with superscript Sh.)

(17) ^{Sh}[hanã^{mb}i]

Now consider the Shizukuishi-chō compound corresponding to Tokyo /kaba+bi/ ‘ceremonial fire’ (cf. /kaba/ ‘birch’, /hi/ ‘fire’), shown in (18). The prenasalisation in the preceding syllable blocks prenasalisation at the *rendaku* site.

(18) ^{Sh}[kã^{mb}abi]

Although the Shizukuishi-chō dialect preserves prenasalised [ᵐb ʳd ʳdz], ^{OJ}/g/ [ᵑg] has become [ŋ], just as in (conservative) Tokyo pronunciation. This [ŋ] does not block prenasalisation in an adjacent syllable, that is, the constraint has maintained its phonetic grounding. For example, compare two compound verbs corresponding to Tokyo /nige+das-u/ ‘run away’ (19a) and /tobi+das-u/ ‘fly away’ (19b).

- (19) a. ^{Sh}[niŋẽndasuw]
b. ^{Sh}[tõ^{mb}idasuw]

If the same constraint holds in the closely related Kahoku-chō dialect, then ^K[nã^{mb}ẽbuda] (16a) is actually the fully nativised form corresponding to Tokyo /nabe+buta/ ‘pot lid’.

Although Miyashita is a proficient speaker of the Kahoku dialect (thanks to spending a lot of time with her grandparents as a child), she does not have the secure native intuitions of Kahoku-chō natives a generation older. As noted in §4, many older Kahoku-chō speakers retain prenasalised ^K[ᵑg]. Does the Kahoku-chō dialect in fact have a phonotactic constraint prohibiting prenasalisation in consecutive syllables? And if so, does ^K[ᵑg] block prenasalisation just like other prenasalised voiced obstruents? Unfortunately, these are questions that the 2012 Kahoku-chō survey did not address. Our fragmentary knowledge at the time did not give us any reason to probe in this direction. Since all northern Tōhoku dialects are endangered, we feel extremely fortunate to have learned of this phonotactic restriction in a closely related dialect, and there is still hope that the necessary follow-up fieldwork in Kahoku-chō can be carried out in the near future.

6. The Domain of OJ Lyman’s Law

As explained in §1, Lyman’s Law in modern Tokyo Japanese blocks *rendaku* when there is a voiced obstruent anywhere in E2. Adjacency is not relevant; a voiced obstruent in E2 blocks *rendaku* even if it is not in the immediately following syllable. Furthermore, a voiced obstruent in the last syllable of E1 does not block *rendaku*, even though it is adjacent to the syllable containing the *rendaku* site.

In northern Tōhoku dialects, on the other hand, the evidence cited in §5 from Shizukuishi-chō indicates that what is prohibited is prenasalised voiced obstruents in adjacent syllables. The strong version of Lyman’s Law in OJ (see §2) was also a constraint involving prenasalised voiced obstruents, and the examples cited in §3, repeated below in (20), are consistent with the claim that adjacency was relevant.

- (20) a. ^{OJ}/yorodu+tabi/ ‘many times’ [jorõndutã^{mb}i] (inferred pronunciation)
cf. ^{OJ}/yorodu/ ‘myriad’, ^{OJ}/tabi/ ‘time’

- b. ^{OJ}/tugi+tugi/ ‘again and again’ [tũ^ɰgitũ^ɰgi] (inferred pronunciation)
 cf. ^{OJ}/tugi/ ‘next’

Each of the examples in (20) contained two prenasalised voiced obstruents, but not in adjacent syllables.

The question of primary interest, of course, is whether the strong version of Lyman’s Law prevented *rendaku* only when *rendaku* would have resulted in prenasalised voiced obstruents in adjacent syllables. The situation in OJ is hard to assess for two reasons. First, the number of phonographically attested compounds is limited. Second, very few OJ morphemes were longer than two syllables.

One important example is given in (21).

- (21) ^{OJ}/sode/ ‘sleeve’ ^{OJ}/tuke/ ‘attaching’ ^{OJ}/koromo/ ‘garment’
^{OJ}/sode+tuke/ ‘sleeved’
^{OJ}/sode+tuke+goromo/ ‘sleeved garment’ [sõⁿdetukẽ^ɰgoromo] (inferred pronunciation)

(21) shows that a prenasalised voiced obstruent not in the morph immediately preceding the *rendaku* site did not block *rendaku*.

It is not clear from (21), however, whether the domain of the inhibiting effect in E1 was the entire preceding morph or just the preceding syllable. Distinguishing these two possibilities requires examples with a monomorphemic E1 that contains a prenasalised voiced obstruent somewhere other than in its final syllable. As noted in §2, word-initial prenasalised voiced obstruents did not occur in the non-mimetic native OJ vocabulary. Consequently, the compounds of interest must have an E1 that was one the few OJ morphs longer than two syllables. The sole relevant example is shown in (22).

- (22) ^{OJ}/madara/ ‘multicolour’ + ^{OJ}/pusuma/ ‘bedding’
^{OJ}/madara+busuma/ ‘multicoloured bedding’ [mãⁿdarã^mbusuma] (inferred pronunciation)

The *rendaku* in (22) indicates that a prenasalised voiced obstruent preceding the *rendaku* site had to be in the immediately preceding syllable to block *rendaku*.

As for E2, unfortunately there are no convincing diagnostic examples involving a prenasalised voiced obstruent that is not in the syllable immediately following the *rendaku* site. The absence of *rendaku* in the examples in (23) is suggestive but by no means conclusive.

- (23) a. ^{OJ}/moto/ ‘origin’ + ^{OJ}/pototogisu/ ‘cuckoo’
^{OJ}/moto+pototogisu/ ‘returned cuckoo’
 b. ^{OJ}/yama/ ‘mountain’ + ^{OJ}/tatibana/ ‘tangerine’
^{OJ}/yama+tatibana/ ‘mountain tangerine’

Both E2s in (23) are etymologically composite, although OJ speakers may not have analyzed them.

For ^{OJ}/pototogisu/ ‘cuckoo’ in (23a), Martin (1987: 416) gives the etymology in (24).

- (24) pre-OJ/poto/ onomatopoetic
 pre-OJ/to/ quotative
 pre-OJ/naki/ ‘crying’
 pre-OJ/su/ ‘bird’
 [pototonakisu] > [pototõ^ɰgisu]

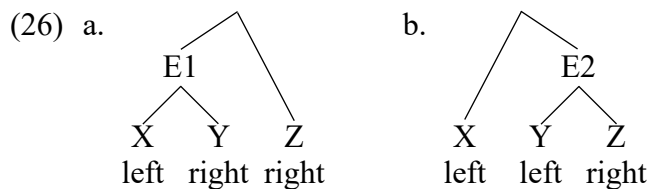
Mimetic morphemes are usually characterised as consistently *rendaku*-immune in modern Japanese (Vance 2015: 416–417). The immunity may not have been quite so consistent in OJ, but distinction between mimetic and non-mimetic is difficult to draw because the boundary

between these two sectors of the vocabulary has always been fuzzy and porous (Hamano 1998:6–7).⁹ In any case, if the first two syllables of ^{OJ}/pototogisu/ derive from a *rendaku*-immune morpheme, there is no reason to attribute the absence of *rendaku* in (23a) to the non-adjacent prenasalised voiced obstruent ^{OJ}/g/.

For ^{OJ}/tatibana/ ‘tangerine’ in (23b), Martin (1987: 543) gives the etymology in (25).

- (25) pre-OJ/ta/ ‘paddy’
 pre-OJ/ti/ ‘path’
 pre-OJ/no/ genitive
 pre-OJ/pana/ ‘flower’
 [tatinopana] > [tati^mbana]

It seems likely that speakers would have seen the morpheme for ‘flower’ in ^{OJ}/tatibana/, and as long as the entire word was felt to be compound of some sort, its resistance to *rendaku* could have developed because of the right-branch condition, which restricts *rendaku* to a right-side branch in constituent structure (Otsu 1980: 217–222; Vance 2007b; Kumagai 2014). The diagrams in (26) illustrate.



In most three-element compounds of the form X+Y+Z, the middle element Y combines with X to form a complex E1, as in (26a), or with Z to form a complex E2, as in (26b). Assuming that neither Y nor Z is immune to *rendaku*, the right-branch condition predicts that *rendaku* is possible in both Y and Z in (26a) but is possible only in Z in (26b). If the pre-OJ ancestor of the word for ‘mountain tangerine’ had the structure ^{pre-OJ}/yama/+{^{pre-OJ}/tati+bana/}, the middle element ^{pre-OJ}/tati/ was on a left branch, like Y in (26b). If the right-branch condition was active in pre-OJ, it would have made any Y+Z combination immune to *rendaku* as an E2, regardless of whether or not there was a (prenasalised) voiced obstruent anywhere in that Y+Z. The upshot is that, originally, the absence of *rendaku* in ^{OJ}/yama+tatibana/ (23b) may have had nothing to do with the ^{OJ}/b/ in ^{OJ}/tatibana/.

The claim that adjacency was crucial for the strong version of Lyman’s Law would be greatly strengthened if there were an example of *rendaku* in a two-element OJ compound with an E2 that (1) was unambiguously monomorphemic, (2) had more than two syllables, and (3) contained a (prenasalised) voiced obstruent in the third syllable or later. Unfortunately, there simply are no such examples.

7. Conclusion

This paper has looked at three kinds of interlocking evidence to argue that Lyman’s Law originated as an OCP-like constraint prohibiting prenasalisation in consecutive syllables. The first kind of evidence comes from typology: constraints on similar consonants in close proximity generally apply not to ordinary voicing but to features with phonetic cues that are more

⁹ The entry in *Nihon kokugo daijiten* (Nihon Kokugo Dai-jiten Dainihan Henshū Iin-kai 2000–02), the Japanese counterpart of the *Oxford English Dictionary*, says that /kira+gira+ši/, with *rendaku*, is attested in the late 11th century, although not in OJ. This is an example of *rendaku* in a base that is mimetic, at least etymologically (cf. modern /kira+kira/ ‘sparkle-sparkle’).

spread out in time, such as aspiration or prenasalisation (§3). The second kind of evidence comes from dialectology: at least some endangered dialects of Japanese with prenasalised voiced obstruents prohibit these marked consonants from occurring in adjacent syllables (§5). The third kind of evidence comes from philology: compounds recorded in phonograms in Old Japanese texts are consistent with a constraint that applies to adjacent syllables (§6).

Needless to say, this account of the strong version of Lyman's Law in Old Japanese does not explain how it morphed into the constraint that we see in modern Tokyo Japanese. Modern Lyman's Law applies not to words but to compound elements (morphemes in the case of native elements), and adjacency does not seem to be relevant. Recent studies on the diachronic development of consonant cooccurrence constraints by Coetzee (2014) and Gallagher (2016) offer some tantalizing hints, but we leave this problem for future research.

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